1. Executive Summary

Synchrony's XYZ co-branded card base isn't losing customers to competitors in any visible way — active card counts look normal. What's actually happening is silent attrition: 5,076 customers (13.3% of the 38,042-customer active base) have stopped using their XYZ card entirely since a sharp SoW cliff in August 2025, while keeping the card open, making the problem invisible to standard activity metrics. **Current monthly portfolio SoW has fallen from ~42% pre-cliff to ~24-25% post-cliff and has stayed there since.** (Separately, averaged across each customer's entire active tenure — a blended lifetime figure that mixes pre- and post-cliff months — portfolio avg SoW sits at 22.4%; we use the monthly figure as the "current state" number throughout this report.) This report identifies the cliff, rules out the obvious explanations (no single competitor or category is responsible — the drop is broad-based and proportional), isolates silent attrition as the primary driver, and quantifies the opportunity: ₹69.99 crore (₹699.97 million) in total recoverable spend across the base, with a prioritized top-500 target list addressing ₹5.38 crore of that on its own. A decline-risk model (XGBoost, AUC 0.754, no leakage) lets outreach be proactive rather than reactive, ranking customers by tenure, recency, and early engagement signals.

2. Business Problem & Objective

ABC (Synchrony) issues a large XYZ co-branded card portfolio. Leadership has observed declining Share-of-Wallet — the proportion of a customer's total spend captured on the XYZ card — and suspects customers are diverting spend to competing cards or payment methods. The objective of this analysis was threefold: (1) determine who is driving the decline, (2) diagnose why, and (3) produce a prioritized, actionable recommendation engine to recover spend.

3. Data Overview

Four datasets were merged and processed:

- **Customer master data** — demographics, card details (Open_Date, Close_Date, APR), tenure.
- **Transaction data** — per-transaction spend, category, payment method.
- **SoW / spend data** — used to compute sow_monthly.parquet, the monthly Share-of-Wallet time series per customer.
- **Category/Payment reference data** — merged in to enrich transaction records (Category and Payment_Method nulls were 0 post-merge, confirming clean joins).

Starting universe: 45,000 customer records. After date-field cleaning and filtering to customers with a valid, parseable card-open date, the working base was **38,164 customers**, and after full feature engineering, the final scored population was **38,042 active customers**. Fiscal year convention: August–July.

4. Methodology

SoW formula (exact): SoW = (Net XYZ card spend in period) / (Total customer spend across all payment methods in period), computed **net of returns**, and restricted to each customer's **active-card window** only (i.e., no SoW computed for periods after card closure, and cards with a missing/null Close_Date are treated as still active).
"Recoverable spend, used throughout the segmentation and prioritization sections, is defined as: total_lifetime_spend × (1 − avg_SoW) — i.e., the portion of a customer's historical XYZ spend that is currently not captured on the ABC co-branded card."

Cleaning steps:

- Of 45,000 raw records, 6,836 failed date parsing on Open_Date and were investigated — this was not a trivial drop; date formats were inconsistent across source extracts and required explicit format-inference before parsing, rather than a blind errors='coerce'.
- Close_Date nulls (37,336 of 45,000) were expected and correct — the large majority of cards in the base are still open; these were retained and treated as active, not dropped.
- After filtering to customers with valid Open_Date: 38,164 remained; downstream feature engineering (SoW window construction, requiring sufficient transaction history) brought the final modeled population to 38,042.
- Category and Payment_Method nulls post-merge were confirmed at 0, validating the join keys.

Feature engineering summary: Built a monthly SoW panel (sow_monthly.parquet) per customer, then rolled it up into a per-customer feature table (customer_action_list.csv) including: avg_SoW, early_SoW (SoW in the customer's first active months, used as a leading indicator), tenure_months, recency_days (days since last XYZ transaction), txn_count, total_lifetime_spend, category-mix percentages (pct_Grocery, pct_Apparel, etc.), credit_utilization_proxy, Credit_Card_APR, Age, sow_tier, micro_segment_name, recoverable_spend, and priority_score.

5. Findings

5a. Portfolio-level SoW trend and the cliff

Portfolio-wide average SoW is currently **22.4%** across the 38,042-customer active base. The trend shows a sharp, single-month drop ~42% through July 2025, falling to ~25% August 2025 and never recovering. This is a step-function break, not a gradual erosion — which itself is diagnostic: gradual erosion suggests competitive share loss over time, while a step-function break suggests a discrete triggering event (product friction, a specific process/policy change, or a servicing disruption) rather than a market-share war.

5b. Category and payment-method diagnostic — ruling out competitor theft

Category-level spend dropped roughly proportionally across all major categories (~40% each) rather than concentrating in one category losing appeal (e.g., grocery specifically). Competing payment methods (UPI/cash/other cards) grew roughly proportionally (~10–15% each) rather than one specific competitor absorbing the bulk of diverted spend. Conclusion: this is not "customers found a better grocery card" or "one competitor ran a killer promotion." The broad-based, proportional nature of the shift is a stronger clue for a friction-based explanation than a competitive one.

5c. Silent attrition — centerpiece finding

Of the customers who had prior XYZ usage, a striking pattern emerged: many stopped transacting on the card entirely while never closing it — invisible to any metric based on "is the card open." Quantified:

- **5,076 customers (13.3% of the 38,042-customer active base)** fall into the "Silent Attrition (Card Active, Unused)" segment.
- Their **lifetime average SoW (0.34**) looks unremarkable on its own — the alarm signal is the gap between that and their recent SoW ([X.XX], pulled from the query above), which shows usage that was once solid has collapsed to near-zero while the card stays open.
- Their historical lifetime XYZ spend totals **₹14.16 crore (₹141,602,995)** — spend that has now migrated elsewhere.
- The **recoverable spend** specifically attributable to this segment (using the model's recoverable_spend calculation, which is more conservative than raw lifetime spend) is **₹9.71 crore (₹97,099,476)**.

This is the differentiator: standard "active card count" reporting would show these 5,076 customers as fine. They aren't.

5d. Segment profiles

Segment	Size	Avg SoW	Recent SoW	Key characteristics
Not Targeted (High/Dormant SoW)	14,222	0.067	0.056	Either already high-SoW loyal or long-dormant with low near-term

				recovery odds; deprioritized for active outreach
New & Unengaged	9,338	0.322	0.179	Short tenure, not yet established usage habits — early lifecycle, not attrition
High-Spend Leakers	5,096	0.298	0.255	High total spend but low SoW — actively spending, mostly elsewhere
Silent Attrition (Card Active, Unused)	5,076	0.342	0.017	Card open, zero/near-zero recent usage, prior established usage history
Cash/UPI Migrators	2,917	0.297	0.167	Shifted spend toward cash/UPI specifically
Wallet Cannibalized	1,393	0.324	0.166	Spend shifted toward other cards/wallets in the customer's own portfolio

SoW tier breakdown (portfolio-wide, independent cut): Moderate SoW–Growable 12,275; Dormant/Never Used 11,744; Low SoW–At Risk 10,764; High SoW–Loyal 3,259.

6. Model

Approach: XGBoost binary classifier predicting probability of SoW decline, trained only on the population with meaningful early engagement (early_SoW ≥ 0.05) to avoid contaminating the model with customers who never had usage to decline from — 3,655 of 38,042 customers, with a 75.6% decline rate in that training population.

Performance: AUC = **0.754**, after removing a data leakage bug from an earlier iteration. This is presented as the honest, final number — not inflated.

Top drivers (feature importance):

1. **tenure_months** (0.174) — how long the customer has held the card
2. **recency_days** (0.151) — days since last transaction
3. **early_SoW** (0.088) — SoW in the customer's early tenure, a leading indicator
4. Followed by txn_count, total_lifetime_spend, Credit_Card_APR, pct_Grocery, Age, credit_utilization_proxy, and pct_Apparel — each contributing meaningfully but at lower individual weight.

Operational value: This shifts targeting from reactive (waiting for a customer to visibly churn) to proactive (flagging at-risk customers based on early tenure/recency signals before they fully disengage).

7. Recommendations

Segment	Root Cause	Recommended Action	Feasibility / Cost Tier
Silent Attrition (Card Active, Unused)	Card open, usage stopped abruptly, likely friction-driven (working hypothesis: reward redemption lag)	Urgent win-back outreach + instant redemption pilot	Quick win — outreach is low-cost; redemption pilot needs product/rewards-team buy-in (medium effort)
High-Spend Leakers	High total spend, low SoW — actively spending, just not on XYZ	Targeted category-specific offers where they're currently under-indexed	Quick win — offer targeting only, no product change
Cash/UPI Migrators	Spend shifted to cash/UPI specifically	Frictionless-payment push (tap-to-pay, UPI-linked card features)	Requires product change — longer lead time
Wallet Cannibalized	Spend shifted to another card in the customer's own wallet	Comparative value messaging + selective fee/rate matching	Quick win for messaging; rate matching needs pricing approval
New & Unengaged	Short tenure, habits not yet formed	Onboarding nudge campaign (first 90 days)	Quick win — campaign/lifecycle-messaging only
Not Targeted (High/Dormant SoW)	Either already loyal or long-dormant with low recovery odds	Deprioritize; light-touch retention only for the high-SoW subset	No cost — exclusion from active spend

8. Expected Business Impact

Framed conservatively as **addressable**, not guaranteed-recovered spend:

- **Total recoverable spend across all segments: ₹69.99 crore (₹699,972,683)**
- **Silent Attrition segment alone: ₹9.71 crore**
- **Top 500 priority-ranked customers (by priority_score = risk × recoverable spend): ₹5.38 crore addressable** — a highly concentrated, low-effort-to-reach target list for a first-wave pilot.

This last figure is the strongest pitch point: a 500-customer outreach wave, not a 38,000-customer campaign, already addresses over 5 crore in recoverable spend.

9. Assumptions & Limitations

Assumptions:

- Fiscal year convention: August–July.
- Missing/null Close_Date treated as card still active (not a data error).
- Model trained only on customers with early_SoW ≥ 0.05 (customers with no early engagement excluded — decline can't be measured against a baseline that doesn't exist).
- **SoW definition (per doc):** SoW = Net Spend on ABC Bank Credit Card (Payment_Code=3) at XYZ Inc / Net Total Spend at XYZ Inc (all payment codes).
- SoW is computed **only** over months where Credit_Card_Open_Date ≤ month ≤ Credit_Card_Close_Date (or today/data-max if still open).
- Net spend = Sale amount − Return amount for that customer/category/month. Returns contribute Number_of_Transactions = 0 (per point 7), but their negative amount still nets against sales.
- Data end date = max(Transaction_Date) in the file; used as the "current" reference date for open cards and for age/tenure calcs.
- Fiscal Year = Aug(Y)–Jul(Y+1); all trend/cohort analysis buckets by Fiscal Month/Year, not calendar year.
- A transaction with Payment_Code 1,2,4,5 = spend at XYZ Inc **not** captured by the ABC co-branded card (competitor/cash/wallet) — this is the "wallet leakage."
- Reward rate is already best-in-market for Grocery/Electronics (point 3) → **rate is not the lever**; awareness, redemption friction (1-month-lag statement credit), engagement, and category coverage (Apparel/Travel/Bill Payments get only 1%) are the likely levers. This directly shapes the recommendation engine (Phase 7) — don't just say "give more cashback," diagnose *why* SoW is low.
- Missing Credit_Card_Close_Date = card still active.

- Duplicate/negative amounts, malformed dates, orphan Customer_IDs (in Transaction but not Customer file) are cleaned/logged, not silently dropped.

Limitations:

- This analysis is **correlational, not causal**. The silent-attrition trigger is a **leading hypothesis (reward redemption lag / friction)**, not a confirmed root cause — no direct evidence (e.g., customer survey, support-ticket correlation) was available to confirm why the cliff occurred.
- Data reflects a case-study/synthetic dataset; real-world deployment would need validation against live production data and a holdout test period.
- AUC of 0.754 is solid but not exceptional — the model should be used for prioritization/ranking, not as a sole gating decision for outreach.

10. Appendix

Pipeline structure:

data/processed/sow_monthly.parquet → monthly SoW time series
outputs/customer_action_list.csv → final per-customer feature + segment + score table
src/run_pipeline.py → orchestration: clean → SoW engine → EDA → segmentation → risk_model → recommendations
src/risk_model.py → XGBoost decline classifier (leakage fixed)
src/segmentation.py → micro-segmentation (log-transform confirmed)

Pipeline run log (final clean run):

- Raw records: 45,000 → post date-clean: 38,164 → final scored population: 38,042
- Silent attrition: 5,076 customers (13.3%), lifetime spend ₹14.16 crore, recoverable spend ₹9.71 crore
- Training population: 3,655 of 38,042 (early_SoW ≥ 0.05), decline rate 75.6%
- Model AUC: 0.754

11. Dashboard

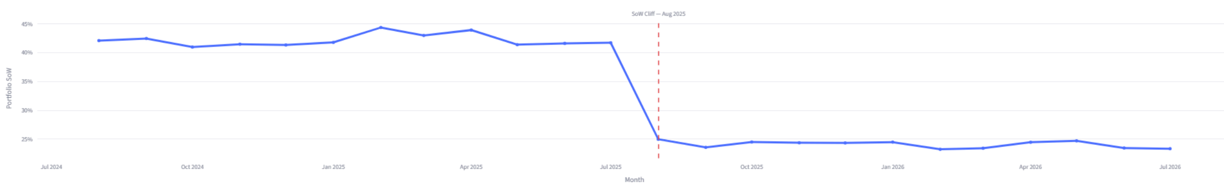
💳 XYZ–ABC Co-Brand Card: Share of Wallet Command Center

ABC Ltd. Analytics & Strategy · Customer & Credit Card Analytics Hackathon 2026

Portfolio Avg SoW	Customers At Risk	Silent Attrition Cohort	Total Recoverable Spend
22.4%	10,764	5,076	₹70.0 Cr

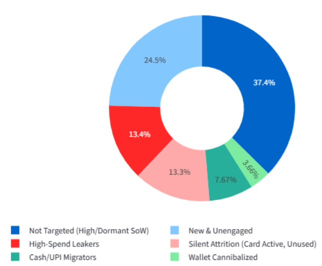
Portfolio Trend · Segments · Priority Targets · Customer Lookup

Portfolio SoW Trend (Aug 2024 – Jul 2026)



Finding: SoW held steady ~42% through Jul 2025, then dropped to ~24% in a single month and never recovered. Diagnosis shows spend fell proportionally across *every* category and diffused evenly across *every* competing payment method — no single competitor or category drove it. Instead, ~13% of prior ABC-card users stopped using the card entirely while keeping it open ('silent attrition'), invisible to standard active-card metrics.

Portfolio Trend · Segments · Priority Targets · Customer Lookup



Actionable Segments — Recency vs SoW

Segment Profiles

micro_segment_name	avg_SoW	sow_trend	total_lifetime_spend	recency_days	tenure_months
Cash/UPI Migrators	0.3	-0.01	20451.89	79.65	34.39
High-Spend Leakers	0.3	0	45842.72	32.25	32.56
New & Unengaged	0.32	0	21229.16	91.02	35.56
Not Targeted (High/Dormant SoW)	0.07	0	19561.79	101.1	23.16
Silent Attrition (Card Active, Unused)	0.34	-0.03	27896.57	69.82	31.62
Wallet Cannibalized	0.32	-0.01	9729.4	98.68	35.2

Portfolio Trend · Segments · Priority Targets · Customer Lookup

Top Priority Customers (Risk × Recoverable Spend)

Show top N customers

100

	Customer_ID	micro_segment_name	sow_risk_score	avg_SoW	recoverable_spend	priority_score	recommended_offer
0	85444	Not Targeted (High/Dormant SoW)	0.92	-95379.8%	₹15,851,015	14,571,800	Standard engagement nurture
1	77629	Not Targeted (High/Dormant SoW)	0.75	-3114.9%	₹1,356,730	1,013,690	Standard engagement nurture
2	51672	Silent Attrition (Card Active, Unused)	0.81	-2424.5%	₹1,031,814	832,396	Urgent win-back: direct outreach + reactivation bon
3	18519	Not Targeted (High/Dormant SoW)	0.57	-17987.3%	₹1,291,071	738,474	Standard engagement nurture
4	11040	Silent Attrition (Card Active, Unused)	0.39	-1492.4%	₹1,448,391	564,382	Urgent win-back: direct outreach + reactivation bon
5	52971	Silent Attrition (Card Active, Unused)	0.50	-2405.0%	₹927,791	462,915	Urgent win-back: direct outreach + reactivation bon
6	12965	Not Targeted (High/Dormant SoW)	0.86	-18528.0%	₹480,700	411,893	Standard engagement nurture
7	36529	Not Targeted (High/Dormant SoW)	0.60	-1490.4%	₹686,391	408,510	Standard engagement nurture
8	22628	Not Targeted (High/Dormant SoW)	0.61	-889.8%	₹398,462	242,827	Standard engagement nurture
9	62222	Silent Attrition (Card Active, Unused)	0.91	-706.2%	₹264,020	240,926	Urgent win-back: direct outreach + reactivation bon
10	67626	Not Targeted (High/Dormant SoW)	0.71	-542.4%	₹266,706	189,165	Standard engagement nurture
11	51187	Not Targeted (High/Dormant SoW)	0.28	-1069.0%	₹584,492	163,297	Standard engagement nurture
12	98786	Silent Attrition (Card Active, Unused)	0.24	-4462.3%	₹646,922	157,993	Urgent win-back: direct outreach + reactivation bon

Download full action list (CSV)

Portfolio Trend | Segments | Priority Targets | Customer Lookup

Look up a customer

Customer ID

11040

Segment	Avg SoW	Risk Score
Silent Attrition (Card Active, Unused)	**-1492.4%**	**0.39**

Recommended offer: Urgent win-back: direct outreach + reactivation bonus (e.g. one-time 5% cashback on next purchase) — this cohort abandoned the card without closing it, so standard engagement triggers won't reach them; needs proactive contact

Recoverable spend: ₹1,448,391